Chapter 5

Representations of finite groups: Further results

5.1. Frobenius-Schur indicator

Suppose that G is a finite group and V is an irreducible representation of G over $\mathbb{C}$.

Definition 5.1.1. We say that V is

- of complex type if $V \not\cong V^*$,
- of real type if V has a nondegenerate symmetric form invariant under G,
- of quaternionic type if V has a nondegenerate skew form invariant under G.

In particular, note that every quaternionic representation is even-dimensional.

Problem 5.1.2. (a) Show that $\operatorname{End}_{\mathbb{R}[G]} V$ is $\mathbb{C}$ for V of complex type, $\operatorname{Mat}_2(\mathbb{R})$ for V of real type, and $\mathbb{H}$ for V of quaternionic type, which motivates the names above.

Hint: Show that the complexification $V_{\mathbb{C}}$ of V decomposes as $V \oplus V^*$. Use this to compute the dimension of $\operatorname{End}_{\mathbb{R}[G]} V$ in all three cases. Using the fact that $\mathbb{C} \subset \operatorname{End}_{\mathbb{R}[G]} V$, prove the result in the